ANALYTICAL CUMULATIVE EXAMINATION

26th October 1996

Last month the Fall 1996 Barringer Lecturer, Dr. Michael J. Gaffey, gave a departmental seminar that discussed the methods and tribulations of what could be called remote sensing analytical chemistry. This highly challenging branch of chemistry deals with the analysis of planetary objects (planets, moons, asteroids) that cannot be directly sampled and can only be viewed at great distances. Several methods have been developed, generally based on the most rudimentary principles of the spectroscopy of inorganic solids.

1. (40%) Figure 1 shows the spectra of light reflected from the surface of four minerals: iron-nickel alloy, olivine ("Ol", $(Fe,Mg)_2SiO_4$), pyroxene ("Px", $(Fe,Mg)SiO_3$) and spinel ("Sp", $MgAl_2O_4$). Describe in the broadest terms (i) the orgins of the features in these spectra and (ii) the overall shape of the curves.

2. (30%) Figure 2 shows a number of spectra obtained from the sunlight reflected from the surfaces of asteroids. Discuss these spectra using (i) the spectra shown in Fig. 1, (ii) the spectra for meteorites shown in Fig. 3.

3. (30%) Discuss any other methods that occur to you for determining the composition of planetary objects that cannot be sampled for laboratory analysis.

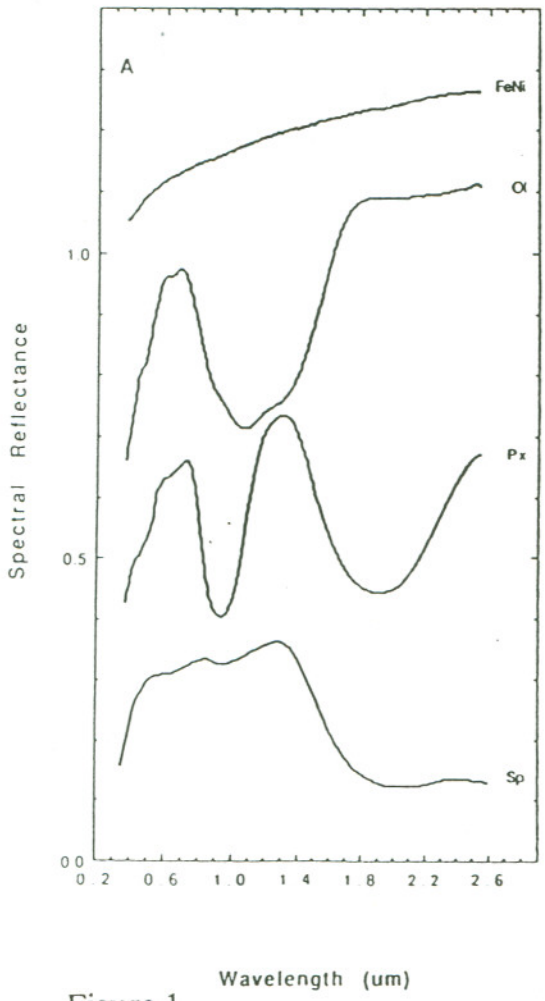


Figure 1.

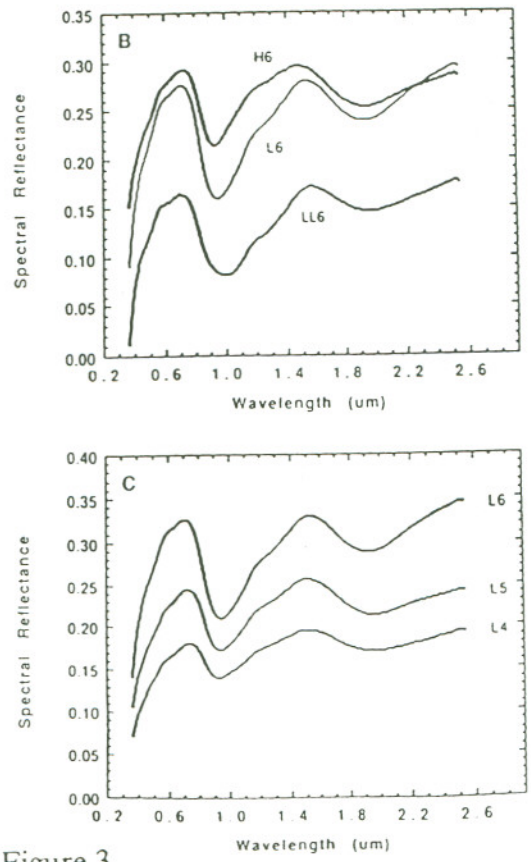
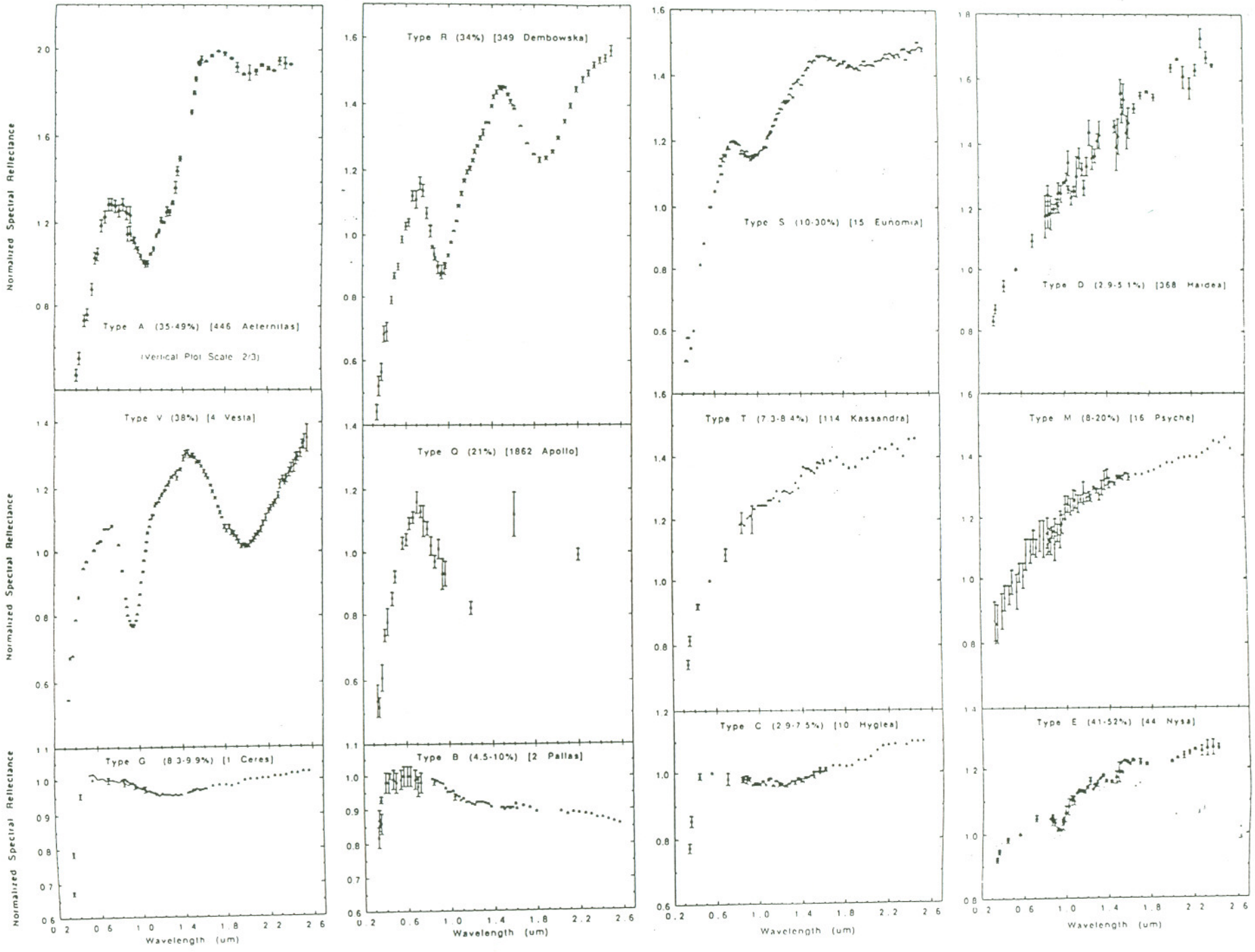


Figure 3.



Asteroid spectroscopy

Figure 2.